

```
1 package project4;
2
3 import java.util.Scanner;
4
5 //Abstract base class
6 abstract class Operation {
7     protected double num1, num2;
8
9     public Operation(double num1, double num2) {
10         this.num1 = num1;
11         this.num2 = num2;
12     }
13
14     public abstract double calculate();
15 }
16
17 //Subclasses for each operation
18 class Addition extends Operation {
19     public Addition(double num1, double num2) {
20         super(num1, num2);
21     }
22
23     public double calculate() {
24         return num1 + num2;
25     }
26 }
27
28 class Subtraction extends Operation {
29     public Subtraction(double num1, double num2) {
30         super(num1, num2);
31     }
32 }
```

```
33 public double calculate() {
34     return num1 - num2;
35 }
36 }
37
38 class Multiplication extends Operation {
39 public Multiplication(double num1, double num2) {
40     super(num1, num2);
41 }
42
43 public double calculate() {
44     return num1 * num2;
45 }
46 }
47
48 class Division extends Operation {
49 public Division(double num1, double num2) {
50     super(num1, num2);
51 }
52
53 public double calculate() {
54     if (num2 == 0) {
55         System.out.println("Error: Cannot divide by zero.");
56         return Double.NaN;
57     }
58     return num1 / num2;
59 }
60 }
61
```

```

62 //Calculator class to handle input/output
63 public class Calculator {
64     public static void main(String[] args) {
65         Scanner scanner = new Scanner(System.in);
66         System.out.println("Simple Calculator");
67
68         System.out.print("Enter the first number: ");
69         double num1 = scanner.nextDouble();
70
71         System.out.print("Enter the second number: ");
72         double num2 = scanner.nextDouble();
73
74         System.out.print("Enter the operation (+, -, *, /): ");
75         char operator = scanner.next().charAt(0);
76
77         Operation operation = getOperation(num1, num2, operator);
78
79         if (operation != null) {
80             double result = operation.calculate();
81             if (!Double.isNaN(result)) {
82                 System.out.println("Result: " + result);
83             }
84         } else {
85             System.out.println("Invalid operation. Please enter a valid operator (+, -, *, /).");
86         }
87
88         scanner.close();
89     }
90
91     public static Operation getOperation(double num1, double num2, char operator) {
92         switch (operator) {
93             case '+':
94                 return new Addition(num1, num2);
95             case '-':
96                 return new Subtraction(num1, num2);
97             case '*':
98                 return new Multiplication(num1, num2);
99             case '/':
100                 return new Division(num1, num2);
101             default:
102                 return null;
103         }
104     }
105 }
106
107

```

Simple Calculator

Enter the first number: 8

Enter the second number: 4

Enter the operation (+, -, *, /): -

Result: 4.0